



23andMe Pharmacogenetics Summary Report

This summary report is intended for healthcare professional use only.

Patient Name: Hannah Mungaia	Ordered By: Hannah Mungaia, self-referral of over-the-counter test	Date Genotyped: 24 Sep 2018
Testing Facility: 23andMe†	Sample Type: Saliva (self-collected and submitted by mail)	Date Printed: 01 Apr 2024

Test Information

23andMe Pharmacogenetics reports provide information about DNA variants that may affect the function of specific proteins involved in the processing of certain medications. This genetic information is intended to help inform discussions with a healthcare professional.

This test does not diagnose any health conditions, provide medical advice, or determine whether a medication is indicated.

Healthcare professionals can learn more at medical.23andme.com/pgt-portal						
GENE ¹	VARIANT(S) DETECTED	TEST RESULT ²	OVERALL FUNCTIONAL EFFECT	THERAPEUTIC AREA	EXAMPLES OF MEDICATIONS PROCESSED IN PART BY THE PROTEIN	MEDICATION INSIGHTS ³
CYP2C19	No variants detected	Predicted normal metabolizer	Normal enzyme function	Cardiology	clopidogrel	Clopidogrel: Likely typical response
				Gastroenterology	omeprazole pantoprazole	
				Infectious Disease	voriconazole	
				Neurology	brivaracetam clobazam	
				Psychiatry	citalopram doxepin escitalopram	Citalopram: Likely typical response
DPYD	No variants detected	Predicted normal metabolizer	Normal enzyme function	Oncology	capecitabine fluorouracil (5-FU)	
SLCO1B1	c.521T>C (one copy)	Predicted decreased function	Decreased protein function	Cardiology	atorvastatin fluvastatin lovastatin pitavastatin pravastatin rosuvastatin simvastatin	Simvastatin: Increased chance of side effects

¹ Variants tested: **CYP2C19**: *2 (c.681G>A, rs4244285), *3 (c.636G>A, rs4986893), *17 (c.-806C>T, rs12248560); **DPYD**: *2A (c.1905+1G>A, rs3918290), D949V (c.2846A>T, rs67376798); **SLCO1B1**: c.521T>C (rs4149056)

² DPYD genotyping results should be confirmed by an independent genetic test prescribed by an individual's own healthcare provider before taking any medical action. SLCO1B1 and CYP2C19 genotyping results (unless the genotype result is CYP2C19 *3/*17 or the metabolizer profile cannot be interpreted) do not require confirmatory testing and may be used by healthcare providers.

³ Medication Insights are currently available for citalopram and clopidogrel (included with the CYP2C19 Drug Metabolism report) and simvastatin (included with the SLCO1B1 Drug Transport report). Interpretive drug information is only available for certain genetic results.

Test Performance

23andMe Pharmacogenetics reports are authorized by the FDA and meet standards of clinical and analytical validity. All saliva samples are processed in CLIA-certified and CAP-accredited labs and are genotyped using a custom version of the Illumina Global Screening Array (GSA). Accuracy of the 23andMe genotyping platform was determined by comparing results from the 23andMe genotyping array with results from sequencing. Greater than 99% of test results were correct. While unlikely, this test may provide false positive or false negative results.

Test Limitations

- Does **not** test for all possible DNA variants in the tested genes that may affect protein function. Having a variant not included in this test may change a person's predicted metabolizer profile or protein function.
- Does **not** test for DNA variants in other genes that may affect other proteins involved in the processing of medications.
- Does **not** provide information on associations between specific DNA variants in the DPYD gene and any specific medications.
- Does **not** account for lifestyle or other health factors that may affect an individual's ability to process medications.

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[†] The 23andMe Personal Genome Service (PGS) is a qualitative genotyping assessment system applied to genomic DNA isolated from human saliva to simultaneously detect, report, and interpret genetic variants in a broad multigene test. The assessment system is intended to enable users to access information about their genetics that could aid discussions with a healthcare professional.

23andMe Pharmacogenetic reports are for over-the-counter use by adults over the age of 18 and provide genetic information to inform discussions with a healthcare professional about processing of therapeutics. The 23andMe Pharmacogenetic report for DPYD describes if a person has variants associated with processing of some therapeutics, but does not describe if a person will or will not respond to a particular therapeutic, and does not describe the association between detected variants and any specific therapeutic. The 23andMe Pharmacogenetic report for CYP2C19 describes if a person has variants associated with processing of some therapeutics and provides interpretive drug information regarding the potential effect of the identified metabolizer phenotype on citalopram and clopidogrel therapy. The 23andMe Pharmacogenetic report for SLC01B1 describes if a person has variants associated with the processing of some therapeutics and provides interpretive drug information regarding the potential effect of the identified transport function phenotype on simvastatin therapy. 23andMe Pharmacogenetic reports are not a substitute for visits to a healthcare professional. The information provided by these reports should not be used by an individual to start, stop, or change any course of treatment. Individuals should consult with their healthcare provider before taking any medical action.